

Sumit Raj

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer (fresher) with an M.Tech in Applied AI and ML from VNIT Nagpur. Built computer vision, deep learning and RAG systems and deployed models with FastAPI and Docker. Seeking an entry-level AI/ML Engineer role.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, CNN, RNN, LSTM, Vision Transformers, Transfer Learning

Computer Vision: OpenCV, YOLOv8, Object Detection, Image Classification, Face Detection, Haar Cascade

GenAI and NLP: LLMs, RAG, LangChain, LangGraph, Hugging Face, Prompt Engineering, Embeddings, LoRA Fine-tuning

MLOps: MLflow, FastAPI, Docker, ONNX, Git, GitHub Actions CI/CD

Math and Data: Pandas, NumPy, Linear Algebra, Probability, Statistics

PROJECTS

Brain Tumor Detection from MRI Scans

2024

PyTorch, Hugging Face, ONNX, Grad-CAM

- Fine-tuned a Vision Transformer on 3,000+ MRI scans: 94.2% accuracy, 0.91 F1-score.
- Used Grad-CAM to visualize model attention and ONNX quantization to cut model size by 60%.

Military Vehicle Detection and Face Authentication Security System

Academic Project

Python, YOLOv8, OpenCV, Haar Cascade

- Built a real-time surveillance system with face authentication at approximately 95% detection accuracy.
- Implemented YOLOv8 military vehicle detection and tracking with automated email alerts for security events.

Mobile Botnet Detection System

2024

Python, TensorFlow, Scikit-learn, SQL

- Analyzed Android network traffic patterns to identify malicious bot communications with C&C servers.
- Engineered 200+ features from network logs and trained a deep neural network with 96.5% precision.

RAG-Based Clinical Question Answering System

2025

Python, LangChain, ChromaDB, FastAPI, Docker, GitHub Actions

- Built a RAG pipeline for clinical document retrieval, reaching 92% answer relevance on a 100-question benchmark.
- Added hybrid search (BM25 + dense embeddings) with Reciprocal Rank Fusion, improving retrieval accuracy by 18%.
- Deployed with FastAPI, Docker and GitHub Actions CI/CD at under 500 ms inference latency.

EXPERIENCE

Software Engineer Intern

Feb 2022 – May 2022

Salesforce (Remote)

- Engineered Python-based bulk data-processing pipelines, reducing manual data-handling effort by approximately 40%.
- Optimized backend workflows and integrated Salesforce APIs, improving system efficiency by approximately 25%.
- Designed secure data-management modules using object-oriented programming; wrote SQL queries (joins, aggregations) for CRM data analysis.

EDUCATION

Visvesvaraya National Institute of Technology (VNIT), Nagpur

2024 – May 2026

M.Tech, Applied Artificial Intelligence and Machine Learning

CGPA: 6.53/10

Dr. D.Y. Patil Institute of Technology, Pune

2019 – 2023

B.E., Computer Science

CGPA: 7.89/10

CERTIFICATIONS

- Coursera: Deep Learning Specialization (in progress)
- DeepLearning.AI: Generative AI with LLMs (in progress)
- Google Cloud: Cloud Fundamentals